

Section 2.3: Document Hierarchy & Structure

Once reading order is established across columns, a raw stream of ordered lines is still insufficient for downstream applications (like ATS parsing or RAG ingestion). To achieve true structural fidelity, the parser must convert isolated lines into semantic document nodes: Headings, Section Titles, Paragraphs, Lists, and Page Anchors.

1. Why Structural Extraction Matters (Information vs. Formatting)

In document extraction, the objective is not visual reproduction, but semantic structure preservation:

- **Flat Text Output:** Loses the distinction between a job title ("Software Engineer"), a company name ("Tech Corp"), and a section header ("WORK EXPERIENCE").
- **Hierarchical Tree Representation:** Maps parent-child relationships so downstream systems know exactly which bullet points belong to which section or job role.

Raw Unstructured Stream	Hierarchical Tree Structure
-----	-----
WORK EXPERIENCE	[Section] WORK EXPERIENCE
Software Engineer, Tech Corp	— [Sub-Header] Software Engineer, Tech Corp
Led migration of microservices.	— [Paragraph] Led migration of microservices.
EDUCATION	— [Section] EDUCATION
B.S. Computer Science	— [Sub-Header] B.S. Computer Science

2. Heading & Section Detection Mechanisms

Engineers use two primary methodologies to detect section boundaries: Deterministic Rule Engines and Visual Layout Models.

A. Deterministic / Font-Attribute Analysis

Native PDF extractors (like pdfplumber or PyMuPDF) inspect low-level font metadata attached to each text span:

- **Font Size & Weight Clustering:**
 - Collect a histogram of all font sizes across the page.
 - Establish the Body Text Baseline Size (typically the statistical mode, e.g., 10pt or 11pt).
 - Any standalone line with font_size > body_baseline_size (e.g., 14pt, 16pt) or is_bold == True is flagged as a candidate header.
- **Spatial Margin & Isolation Signals:**
 - *Vertical Whitespace (Leading):* Headings usually feature larger top margins (margin_top > 1.5 * line_height) than bottom margins.
 - *Horizontal Width & Standalone Check:* Section headers rarely span an entire line; they

are short phrases ending without punctuation like periods (.) or commas (,).

- **Text Case & Keyword Matching:**

- Regex matching for standard uppercase headers (WORK EXPERIENCE, SUMMARY, EDUCATION, PROJECTS) or Title Case patterns.

B. Deep Learning / Vision Layout Parsing

Commercial APIs (Adobe PDF Extract, Google Document AI Layout Parser) and ML tools (Marker) render the page to detect bounding boxes visually. They classify text blocks using categories like Title, Section-Header, Sub-Header, Body-Text, and Caption regardless of underlying font metadata.

3. Paragraph Grouping & Line-Break Consolidation

PDFs do not store paragraphs—they store individual text positioning operations (TJ / Tj operators). Merging lines into continuous paragraphs requires calculating spatial gaps and sentence grammar.

A. Spatial Gap Analysis

- **Line Height (Leading):** Measure vertical distance between baselines:

$$\Delta Y = Y_{\text{line } i} - Y_{\text{line } i+1}$$

- **Threshold Rules:**

- If $\Delta Y \approx \text{standard line height}$ (e.g., $1.2 \times \text{font_size}$), line $i + 1$ continues the current paragraph.
- If $\Delta Y > 1.5 \times \text{line height}$, a hard paragraph boundary is inserted.

B. Linguistic & Grammatical Continuation Rules

Because layout spacing in custom PDFs can be inconsistent, spatial rules are combined with textual heuristics:

Line i : "Developed a distributed microservice architecture using Python and"

Line $i+1$: "Go to handle over 10,000 requests per second."

- **Hyphenation Restoration:** If Line i ends with a trailing hyphen ("and-"), strip the hyphen and merge with Line $i + 1$ ("and Go").
- **Sentence Termination Check:** If Line i ends with a lowercase letter, comma, or conjunction, and Line $i + 1$ starts with a lowercase letter, force a soft line break (space)

instead of a paragraph break.

4. List Structure & Bullet Point Preservation

List items (job responsibility bullets, skill lists) require special handling to preserve indentation depth and nested relationships.

Level 0: • Senior Software Engineer

Level 1: - Led backend team

Level 2: * Reduced latency by 45%

A. Bullet & Symbol Identification

Parsers scan line prefixes for list indicators:

- **Standard Unicode Bullets:** • (\u2022), ▪ (\u25aa), ◦ (\u25e6), - (\u2023).
- **ASCII Bullet Substitutes:** -, *, >.
- **Numbered Lists:** Regular expressions for 1., 1), (a), i., 13..

B. Spatial Indentation & Hanging Indents

- **First-Line Offset:** The bullet character sits at horizontal position X_{bullet} .
- **Hanging Indent Alignment:** Wrapped text on subsequent lines aligns at $X_{\text{text}} > X_{\text{bullet}}$.
- **Nesting Depth Calculation:** The horizontal offset $\Delta X = X_{\text{bullet}} - X_{\text{margin}}$ determines the list nesting level:

$$\text{Depth Level} = \left\lfloor \frac{\Delta X}{\text{Indent Step (e.g., 12pt)}} \right\rfloor$$

5. Representing Document Hierarchy in Output Formats

To pass structured content to downstream processors or LLMs, the document is serialized into JSON Schema or Structured Markdown.

A. Structured JSON Serialization

```
{  
  "document": "Resume_Jane_Doe.pdf",  
  "sections": [  
    {  
      "section_title": "WORK EXPERIENCE",  
      "level": 1,  
    }  
  ]  
}
```

```

"page": 1,
"bounding_box": [50.0, 200.0, 550.0, 215.0],
"blocks": [
  {
    "type": "sub_header",
    "text": "Senior Backend Engineer | Cloud Systems Inc.",
    "metadata": { "font_size": 12.0, "is_bold": true }
  },
  {
    "type": "list_item",
    "depth": 0,
    "bullet_symbol": "•",
    "text": "Architected low-latency ingestion engine handling 1.5M daily events."
  }
]
}

```

B. Clean Markdown Representation

Markdown is the ideal format for LLM ingestion because headers (#, ##), lists (-), and bold text (**) preserve document structure with minimal token overhead:

WORK EXPERIENCE

Senior Backend Engineer | Cloud Systems Inc.

* Architected low-latency ingestion engine handling **1.5M daily events**.

* Reduced database query times by 40% through indexing optimization.

6. Edge Cases & Failure Modes in Structural Parsing

- **Inline / Horizontal Headers:** Headers placed on the same baseline as content (e.g., "Skills: Python, Java, C++"). Requires regex prefix splitting rather than block segmentation.
- **Fake Bullets (Vector Graphics):** Bullet points drawn as vector circles/squares rather than font characters. Native text tools see only spaces, dropping the bullet entirely unless vector shape extraction is enabled.
- **Inconsistent Font Metadata:** In PDFs converted from Canva or HTML-to-PDF, every text block might use the same font size (e.g., 12pt) with visual bolding achieved via duplicated overlapping text vectors, breaking font-size-based heading algorithms.